



Sunbeam Institute of Information Technology
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Module - Concepts of Operating Systems and Administration

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- A **user** is an entity that can log into the system and execute commands.
- Every user has:
 - ✓ Username
 - ✓ User ID (UID)
 - ✓ Home directory
 - ✓ Default shell
- A **group** is a collection of users.
- Used for:
 - ✓ Managing permissions
 - ✓ Controlling access to files/resources

- **Why Users & Groups are Important?**
 - ✓ Security (restrict access)
 - ✓ Multi-user environment
 - ✓ Resource sharing
 - ✓ Permission management
- Linux is a **multi-user system**, so:
 - Multiple users can work simultaneously
 - Each user has controlled access
- Every user and group is uniquely identified by User ID(UID) and Group ID(GID) respectively.

- **Super user (Root user) :**

- The administrator of the system
- Has full control over everything
- Root user has username as root and UID as 0
- Permissions
 - Can read, write, execute any file
 - Can create/delete users
 - Can modify system configuration

- **System users :**

- Used by system services and processes (daemons)
- Not meant for human login
- Usually:
 - UID < 1000 (in most Linux systems)
 - No login shell or limited shell
- Run background services like Web server, Database, Daemons

- **Regular users (Normal users) :**

- Real human users
- Used for daily work
- UID ≥ 1000 (generally)
- Every user has:
 - Home directory (/home/user)
 - Login shell (/bin/bash)
- Permissions
 - Limited access
 - Cannot modify system files

- Stores basic user information (Readable by all users)
- Every user is represented by a row having 7 columns

username:x:UID:GID:comment:home_directory:shell

- devendra:x:1001:1001:Devendra Dhande:/home/devendra:/bin/bash

Field	Meaning
username	Login name
x	Password placeholder (actual password in /etc/shadow)
UID	User ID
GID	Primary Group ID
comment	Full name / description
home_directory	User's home directory
shell	Default login shell

- Stores encrypted passwords
- Only root can read

username:encrypted_password:last_change:min:max:warn:inactive:expire

- devendra:\$6\$abc123\$xyz...:19800:0:99999:7::

Field	Meaning
username	Login name
encrypted_password	Hashed password
last_change	Days since password changed
min	Minimum days before change
max	Maximum days before expiry
warn	Warning days before expiry
inactive	Days after expiry to disable
expire	Account expiry date

! - Account locked
* - Login disabled
empty - No password

- Stores group information

group_name:x:GID:user_list

- developers:x:1001:devendra,alice,bob

Field	Meaning
group_name	Name of group
x	Password placeholder
GID	Group ID
user_list	Members of group

- Users can belong to multiple groups
- Primary group → defined in /etc/passwd
- Secondary groups → listed here

- /etc/passwd is **publicly readable**
- /etc/shadow is **restricted**
- Passwords are **hashed, not stored in plain text**
- UID & GID are used internally, not usernames

- **command : useradd [options] username**

Option	Meaning
-u UID	Set user ID
-g GID	Set primary group
-G groups	Set secondary groups
-d dir	Home directory
-m	Create home directory
-s shell	Login shell
-c "comment"	Description
-e date	Account expiry
-f days	Inactive days after expiry

- Without -m, home directory may not be created
- Default settings come from **/etc/default/useradd**

- **command : usermod [options] username**

Option	Meaning
-u UID	Change UID
-g GID	Change primary group
-G groups	Set secondary groups
-aG groups	Append to groups (very important ★)
-d dir	Change home directory
-m	Move home directory
-s shell	Change shell
-c "comment"	Modify description
-l newname	Change username
-L	Lock account
-U	Unlock account
-e date	Set expiry

- **command : userdel [options] username**

Option	Meaning
-r	Remove home directory
-f	Force delete

- useradd → creates entry in:
 - /etc/passwd
 - /etc/shadow
 - /etc/group
- usermod → updates these files
- userdel → removes entries

- Password complexity can be set using file `/etc/security/pwquality.conf`
- **passwd:**
 - Users can change their passwords using `passwd` command
 - As the root user, you can change a password for any account.
 - > sudo passwd <username>
 - It works with following parameters
 - -d: Delete a password and disable the account
 - -e: Immediately expire a password, forcing a password change by the user
 - -l: Lock the account (for example, during a leave of absence)
 - -u: Unlock a locked account
- **chage:**
 - Password requirements are also configured by using the `chage` command
 - It works with following parameters
 - -l: shows the current values configured for the user
 - -M: sets maximum number of days between password change
 - -m: sets minimum number of days between password change
 - -W: sets number of warning days before password expires
 - -E: lock an account after specified date

- A group is a collection of users that need to share access to files and other system resources
 - Groups can be used to grant access to files to a set of users instead of just a single user
 - Like users, groups have group names to make them easier to work with
 - Internally, the system identifies groups by the unique identification number known as group ID or GID
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- **Types**
 - **Primary Group**
 - Every user has exactly one primary group
 - By default this is the group that will own new files created by the user
 - Normally, when you create a new user, Linux adds a new group with the same name
 - **Supplementary Groups**
 - User may be a member of one or more supplementary groups
 - Membership is determined by /etc/group
 - Users are granted access based on whether any of their groups have access

- Create group - groupadd [options] groupname

Option	Meaning
-g GID	Set group ID
-r	Create system group
-f	Force (ignore if exists)
-K KEY=VALUE	Override defaults

- Modify group - groupmod [options] groupname

Option	Meaning
-g GID	Change group ID
-n newname	Rename group

- Delete group - groupdel groupname

- Cannot delete a group if it is a Primary group of a user
- Must change user's primary group first

- Groups help manage permissions efficiently
- Users can belong to:
 - One primary group
 - Multiple secondary groups
- Always use : usermod -aG

- As a sysadmin, you can use a few different files to set the system up the way your institution prefers
- Use /etc/profile to set system-wide environment variables and startup programs for new user shells
- Use /etc/bashrc to establish system-wide functions and aliases for new user shells
- The ~/.bash_profile sets user-specific environment variables for new Bash shells, and ~/.bashrc runs when noninteractive shells are launched
- The tilde character (~) represents the current user's home directory
- The system-wide files process first, and then the user-specific files are executed
- The user-specific configuration files take precedence over system files, allowing users to customize their environments to suit their needs





